

Sample Notes — The Virelia Star System

The **Virelia Star System** is a fictional planetary system centered around the star **Virelia**. Four planets have been discovered orbiting the star: **Asteron, Belara, Cyrene, and Doros**.

Asteron

Asteron is the **closest planet to Virelia**. It is a small, rocky planet with a very thin atmosphere.

Because Asteron is located close to its star, its surface experiences extremely high temperatures during the day.

Asteron completes one orbit around Virelia every **42 Earth days**.

Belara

Belara is the **second planet** from Virelia.

It has a thick atmosphere and large amounts of **liquid water** on its surface. Because of these conditions, scientists consider Belara the planet in the system **most likely to support life**.

Belara has **two known moons**.

Cyrene

Cyrene is the **largest planet** in the Virelia System.

It is a gas planet surrounded by a thin ring system made primarily of **ice and rock**.

Cyrene has **11 known moons** and completes one orbit around Virelia every **8 Earth years**.

Doros

Doros is the **farthest known planet from Virelia**.

It is an icy planet with a very thin atmosphere. Because of its distance from the star, Doros has the **lowest average surface temperature** of the four known planets.

Doros has **three known moons**.

The Nox Black Hole

A black hole called **Nox** is located outside the main planetary region of the Virelia System.

Nox formed after the collapse of a **massive star**.

The boundary surrounding the black hole is called the **event horizon**. Once matter crosses the event horizon, it cannot return to the region outside the black hole.

Scientists cannot directly observe Nox itself. Instead, they study its effects on nearby matter.

Gas and dust moving toward Nox can form a rapidly rotating structure called an **accretion disk**.

As material in the accretion disk moves around the black hole, it can become extremely hot and produce large amounts of radiation.

Key Facts

- Asteron is closest to Virelia.
- Belara is considered the most likely planet to support life.
- Cyrene is the largest planet.
- Doros is the farthest known planet from Virelia.
- Cyrene has the most known moons.
- Nox is a black hole outside the main planetary region.
- Matter that crosses Nox's event horizon cannot return to the outside region.
- An accretion disk can form from gas and dust moving toward Nox.